Advanced Methods: Transforms (MAST90067)

Graduate coursework / Points: 12.5 / Not available in 2025

You're viewing the 2025 Handbook:

2025

Or view archived Handbooks

View full page

About this subject

- Overview
- Eligibility and requirements
- Assessment
- Dates and times
- Further information

Overview

Fees	Look up fees

This subject develops the mathematical methods of applied mathematics and mathematical physics with an emphasis on integral transform and related techniques. An introduction is given to the calculus of variations and the Euler-Lagrange equation. Advanced complex contour integration techniques are used to evaluate and invert Fourier and Laplace transforms. The general theory includes convolutions, Green's functions and generalized functions. The methods of Laplace, stationary phase, steepest descents and Watson's lemma are used to asymptotically approximate integrals. Throughout, the theory is set in the context of examples from applied mathematics and mathematical physics such as the brachistochrone problem, Fraunhofer diffraction, Dirac delta function, heat equation and diffusion.

2024 Q4: Topic 2 (power series and convergence)

Question 4 (12 marks)

Use power series to solve for $y(x)$ around $x = 0$, given

$$(x^2 + 1)y'' - 4xy' + 6y = 0.$$

2024 Q3: Topic 3 (Laplace transforms)

Question 2 (12 marks)

Consider the IVP

$$x''(t) + 2x'(t) + x(t) = 0 \qquad \text{where} \qquad x(0) = a,\ x'(0) = b$$

(a) Solve the IVP using the method of Laplace transforms.

Theorem (Higher order derivatives)

Let $f, f', \ldots, f^{(n-1)}$ be continuous on $[0, \infty)$ and have Laplace transforms. Then

$$\mathscr{L}\left\{f^{(n)}\right\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \ldots - f^{(n-1)}(0)$$

Can be proved using integration by parts and induction.

We will use the case of a second derivative extensively:

$$\mathscr{L}\{f''\} = s^2 F(s) - s f(0) - f'(0)$$

MAST20030 Differential Equations Formulae Sheet

1) Laplace Transforms

1.	$f(t)$	$F(s) = \mathscr{L}(f) = \int_0^\infty f(t)e^{-st}\,dt$ (Definition of Transform)
2.	1	$\frac{1}{s}$
3.	t^n	$\frac{n!}{s^{n+1}}$
4.	e^{at}	$\frac{1}{s-a}$
5.	$\sin(at)$	$\frac{a}{s^2+a^2}$
6.	$\cos(at)$	$\frac{s}{s^2+a^2}$
7.	$\sinh(at)$	$\frac{a}{s^2-a^2}$
8.	$\cosh(at)$	$\frac{s}{s^2-a^2}$

Exercise 3.13: Find the Laplace transform of $\mathrm{e}^{3\,t}\,t^2$

Theorem (s-shift)

If $\mathscr{L}\{f\}$ exists for $\mathrm{Re}(s) > \alpha$, then

$$\mathscr{L}\{\mathrm{e}^{\omega\,t} f(t)\} = F(s-\omega), \qquad \mathrm{Re}(s) > \alpha + \omega$$

$$\mathscr{L}\{t^2\} = \frac{2}{s^3} \quad \text{for} \quad \mathrm{Re}(s) > 0$$

$$\Rightarrow \quad \mathscr{L}\{\mathrm{e}^{3t}\,t^2\} = \frac{2}{(s-3)^3} \quad \text{for} \quad \mathrm{Re}(s) > 3$$

(c) Find the Laplace transform of

$$\int_0^t \delta(\tau - 2) d\tau$$

where $\delta(t)$ is the dirac-delta 'function'.

The Dirac delta, $\delta(t)$

Definition

We define $\delta(t)$ such that

1. $\delta(t) = 0$ for all $t \neq 0$,
2. $\displaystyle\int_{-\infty}^{\infty} \delta(t)\mathrm{d}t = \int_{0}^{a} \delta(t)\mathrm{d}t = 1$ for all $a > 0$, and
3. $\mathscr{L}\{\delta(t)\} = 1$.

- $\delta(t)$ is not a function in the traditional sense (it is a *generalised function* as you will discover in later courses)
- There are various forms of $\delta(t)$ with others being centred at or left of zero; we will always use the above definition

(d) Use the method of Laplace transforms to find $y(t)$ given

$$\int_0^t \delta(\tau - 2)d\tau = 5\int_0^t y(\tau)d\tau + y'(t) + 2y(t), \qquad \text{where} \qquad y(0) = 0.$$

Exercise 3.15: Find the Laplace transform of $f = \sin(t-3)H(t-3)$

Theorem (t-shift)

If $F(s)$ exists for $\mathrm{Re}(s) > \alpha$, and $c \geqslant 0$, then

$$\mathscr{L}\Big\{ f(t-c)H(t-c) \Big\} = \mathrm{e}^{-cs} F(s), \quad \mathrm{Re}(s) > \alpha$$

$$F(s) = \mathrm{e}^{-3s} \mathscr{L}(\sin t)$$

$$\Rightarrow \quad F(s) = \frac{\mathrm{e}^{-3s}}{s^2+1} \quad \text{for} \quad \mathrm{Re}(s) > 0$$

Exercise 3.22: Solve the following integro-differential equation

$$y'(t) + 4\int_0^t y(\tau)\mathrm{d}\tau = 0 \qquad \text{where} \qquad y(0) = 3$$

Laplace transform of an antiderivative

$$\mathscr{L}\left(\int_0^t f(\tau)\mathrm{d}\tau\right) = \mathscr{L}(1 * f) = \mathscr{L}(1)F(s) = \frac{F(s)}{s}$$

Taking the Laplace transform gives

$$\begin{aligned} sY - 3 + 4Y * \mathscr{L}(1) = 0 \quad &\Rightarrow \quad sY - 3 + \frac{4Y}{s} = 0 \\ s^2Y - 3s + 4Y = 0 \quad &\Rightarrow \quad (s^2+4)Y = 3s \\ Y = \frac{3s}{s^2+4} \quad &\Rightarrow \quad Y = \mathscr{L}^{-1}\left(\frac{3s}{s^2+4}\right) = 3\cos 2t \end{aligned}$$